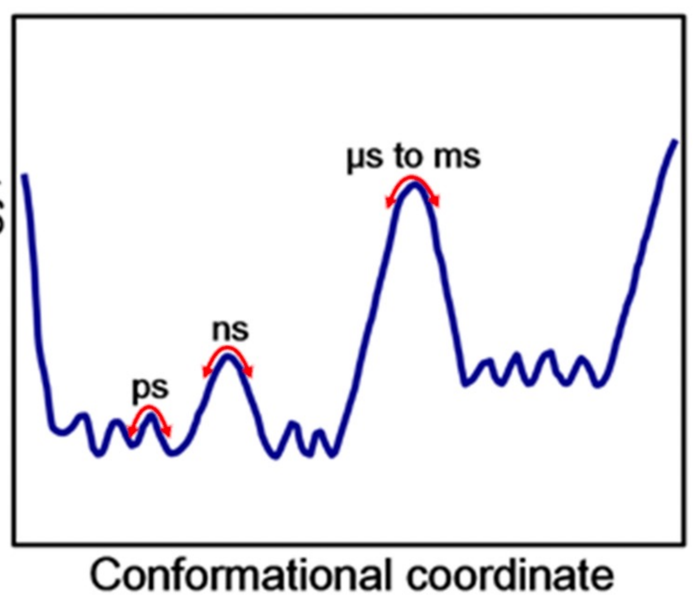
COMPARISON OF ADVANCED SAMPLING TECHNIQUES FOR ATOMISTIC SIMULATION OF RNA FOLDING

Kosar Rahimi[1], Pablo M Piaggi[2], Gül H Zerze[1]

1. Department of Chemical and Biomolecular Engineering, University of Houston
2. Department of Chemistry, Princeton University

Background and motivation

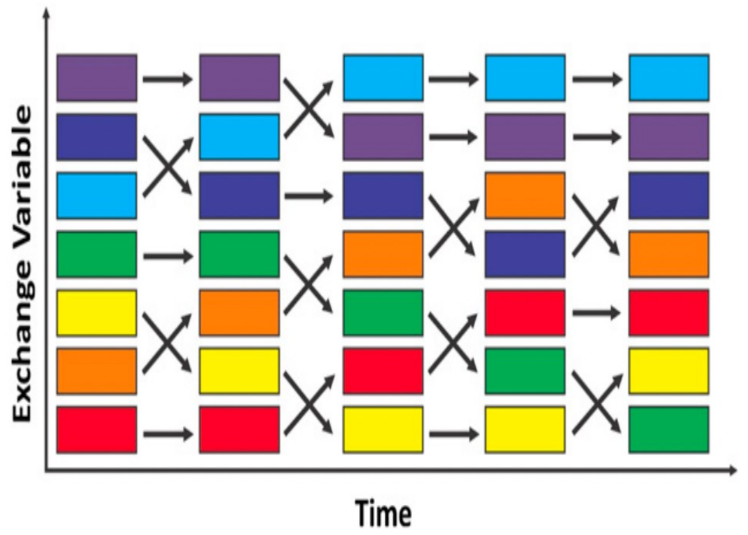


Henzler-Wildman and Kern, 2007, *Nature*, *450*(7172)

>> The computational cost for the plain simulation of complicated biomolecular systems with rough energy landscapes may not be affordable.

>> Advanced sampling helps to facilitate the crossing of energy barriers.

>> Tempering based sampling

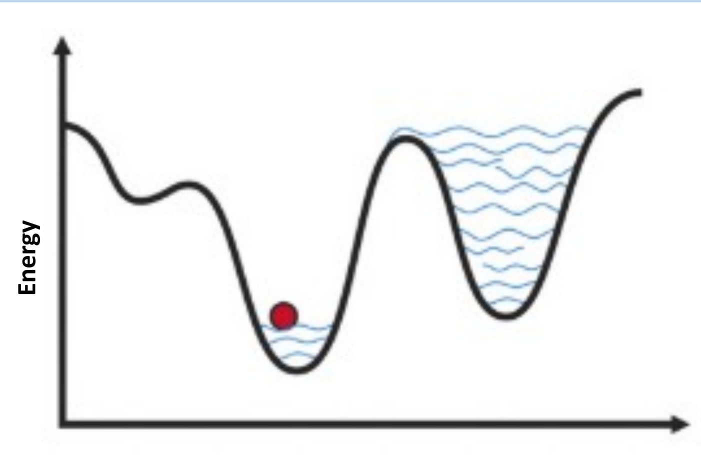


Bernardi et al., 2015, *BBA-General Subjects 1850*(5)

>> Multiple replicas are simulated at different temperatures and a Monte Carlo acceptance criterion is used for coordinate exchange.

⊖ These sampling methods like replica exchange cannot help to overcome more than a few KTs.

>> Collective variable based sampling



Bernardi et al., 2015, *BBA-General Subjects 1850*(5)

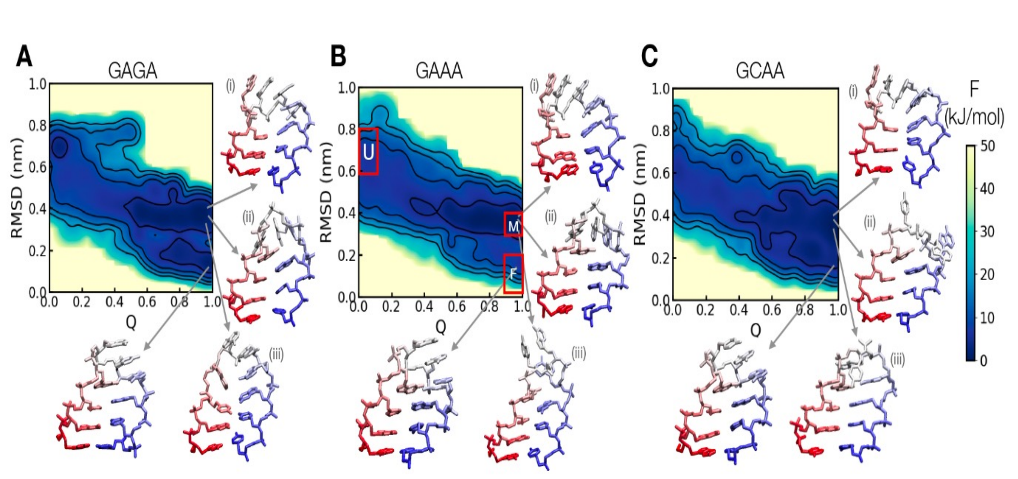
>> Appropriate collective variables (CVs) are chosen and bias potentials as a function of them will be added.

⊖ Many systems are too complex that not all degrees of freedom can be reduced to collective variables.

>> Sampling method used to study RNA tetraloop folding: PTWTE-WTM

✓ Not leaving any DOF ← Parallel-Tempering in the Well-Tempered Ensemble combined with

✓ Overcoming large barriers ← Well-Tempered Metadynamics



Zerze et al., 2021, *JPCB 125*(50)

✓ Folded, misfolded, and unfolded configurations of RNA tetraloops were successfully sampled.

⊖ High computational cost

Simulation method

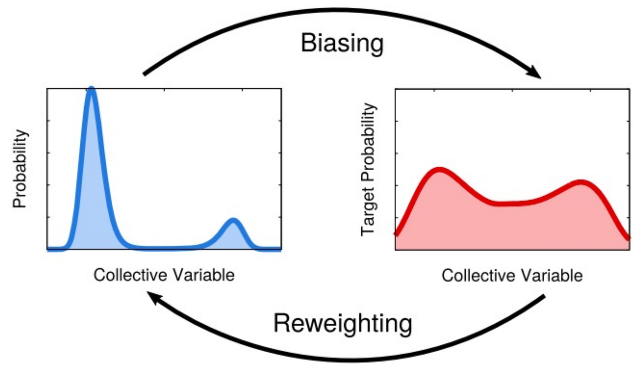
>> The single-stranded GNRA tetraloop of GGCGAGAGCC (GAGA) was simulated.

>> Unfolded initial coordinates of the tetraloops were generated using the Nucleic Acid Builder tool of AmberTools.

>> A combination of a nucleic acid force field and a water model, DESRES ff and TIP4PD, were employed as force fields.

>> Multithermal-Multiumbrella On-the-fly Probability Enhanced Sampling (MM-OPES) was used for sampling the tetraloop folding.

Can MM-OPES **replace** PTWTE-WTM and be **more efficient**?

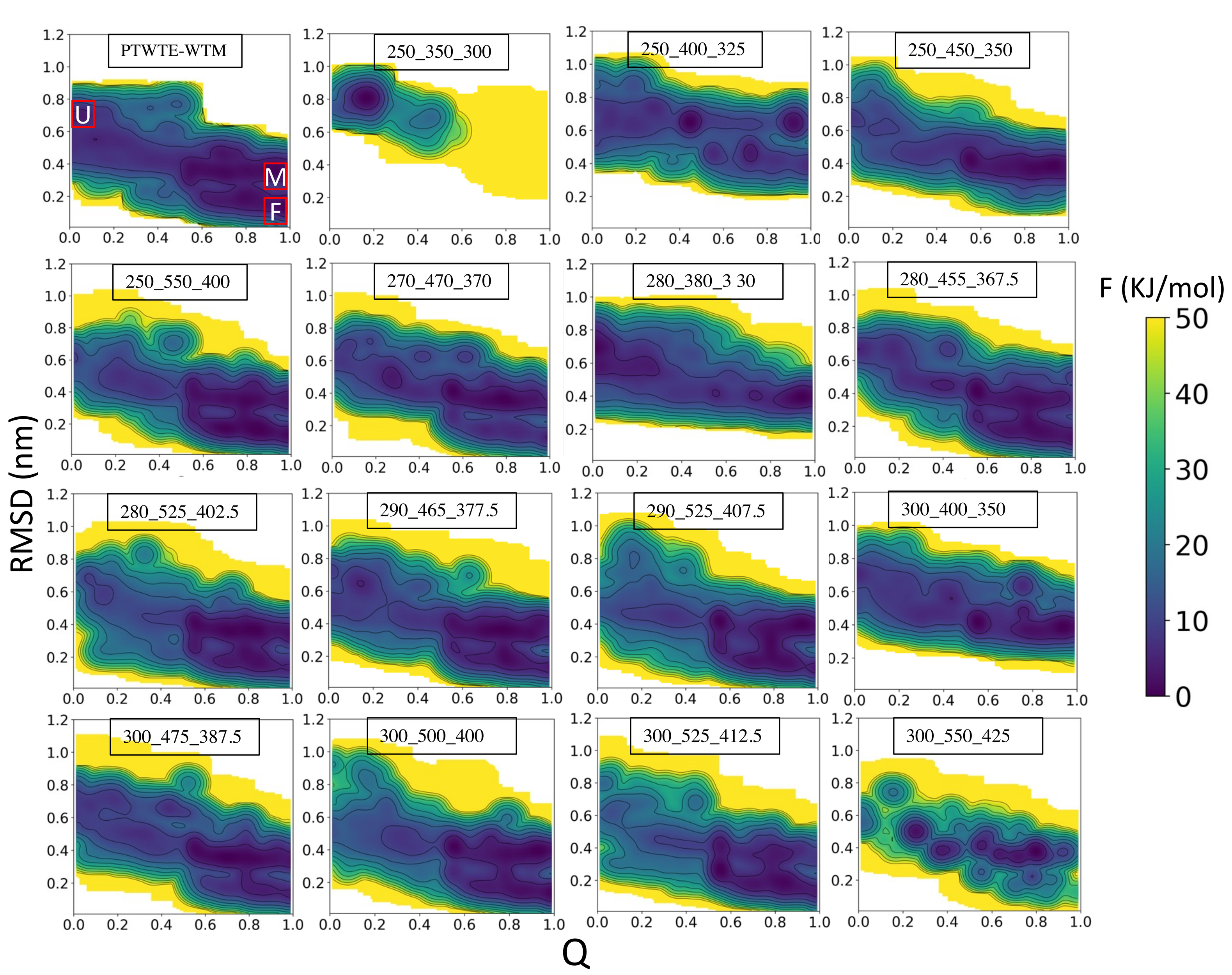


Invernizzi, M., & Parrinello, M.,2020, *J. Phys. Chem*, *11*(7)

Results and discussion

>> Free energy surface of tetraloops at 300 K for simulations using PTWTE-WTM and MM-OPES at different temperature ranges.

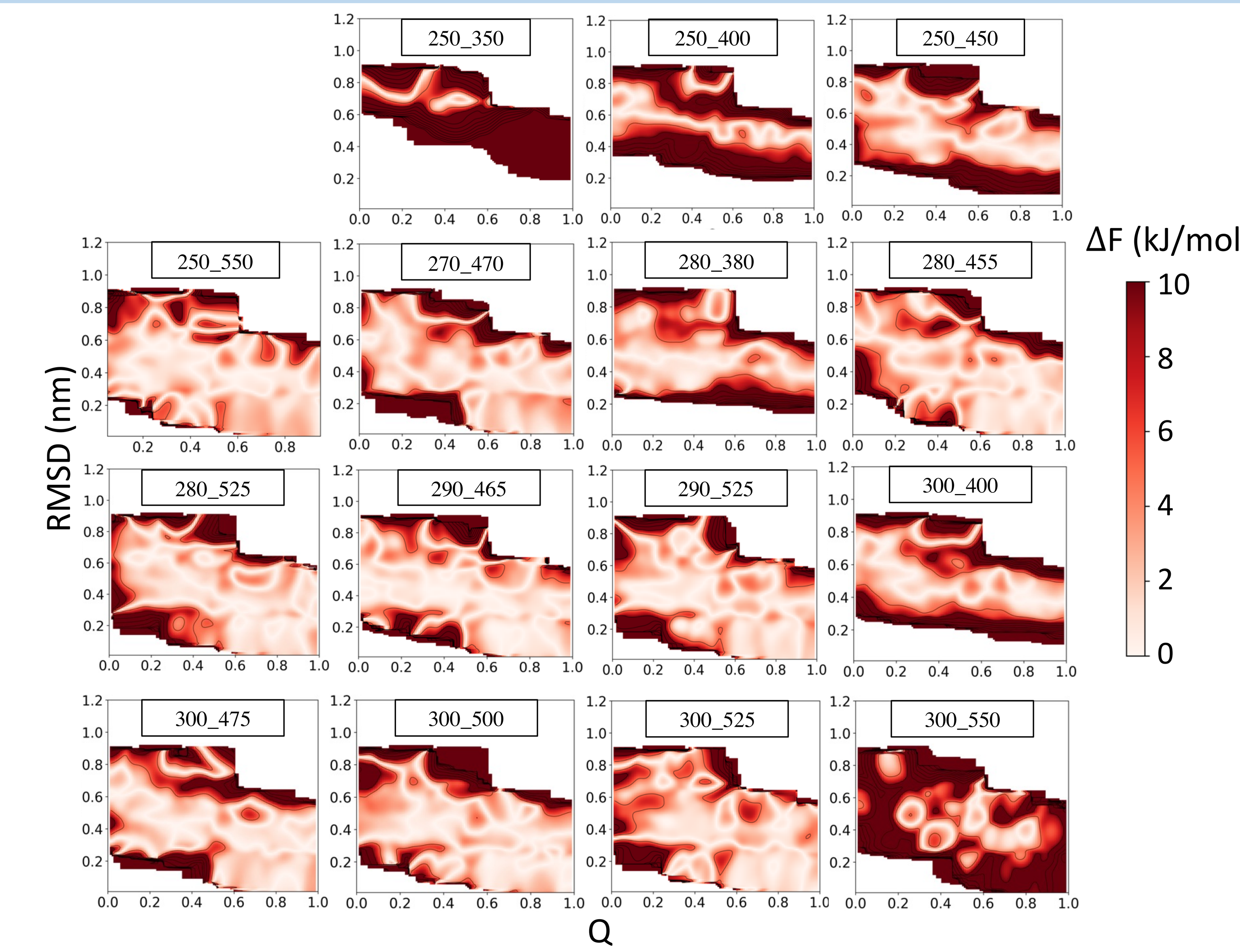
>> At what temperature ranges are RNA configurations sampled ideally?



(unfolded, $Q < 0.11$, $0.6 < \text{RMSD} < 0.8$), (correctly folded, $Q > 0.9$, $\text{RMSD} < 0.2$), (misfolded, $Q > 0.9$, $0.3 < \text{RMSD} < 0.4$)

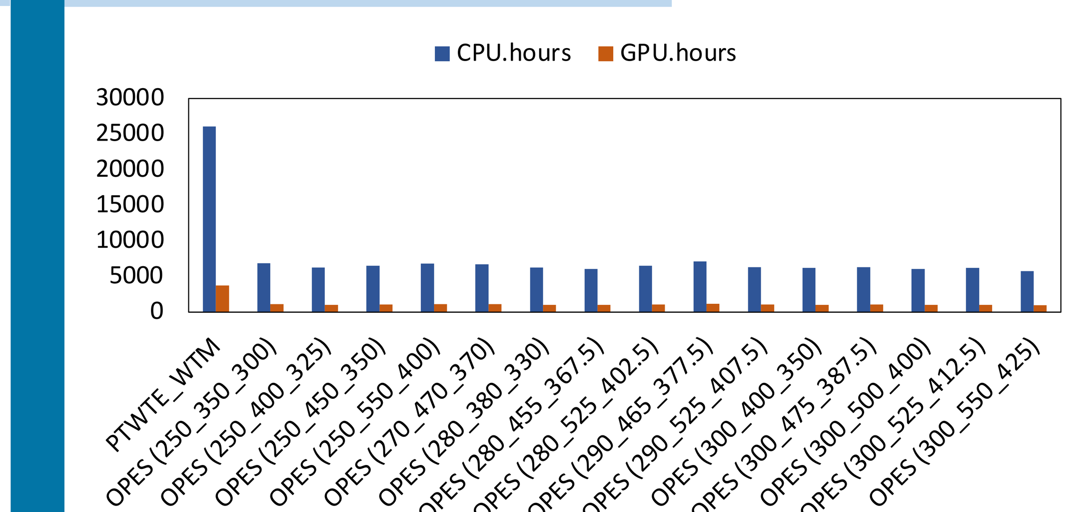
>> Folded, unfolded, and misfolded configurations can be detected in most MM-OPES simulations, similar to PTWTE-WTM.

>> Difference in free energy when using MM-OPES and PTWTE-WTM



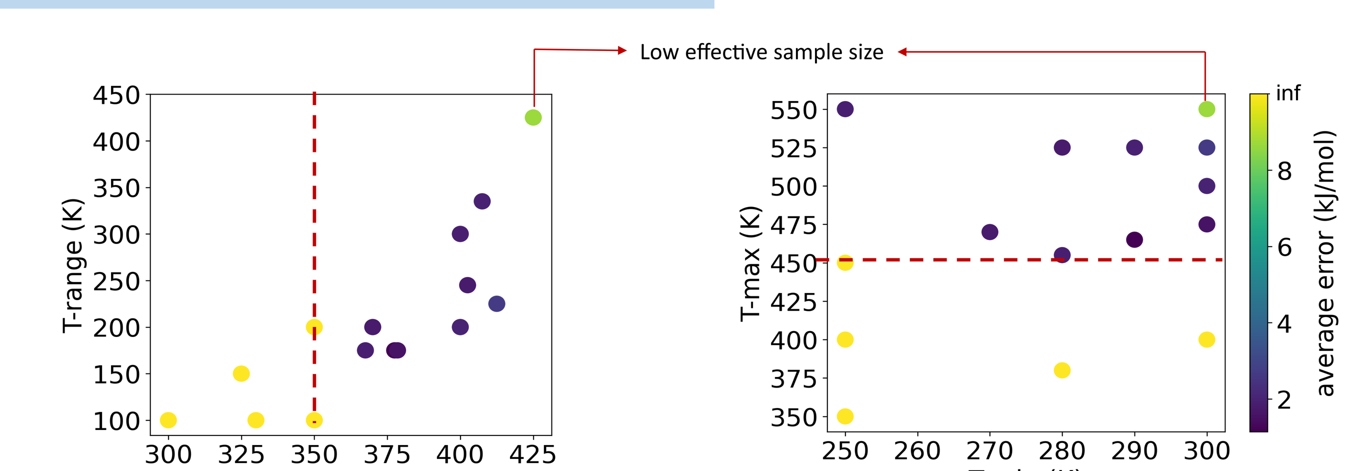
>> low difference in free energy of folded, misfolded, and unfolded basins of most MM-OPES simulations can show the reliability of this method.

>> Computational cost

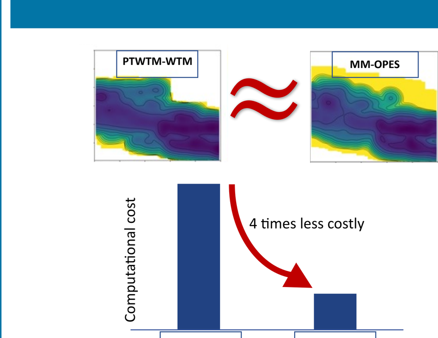


>> The simulation using MM-OPES employed **CPU.hours and GPU.hours less than one-forth** of the ones in the simulation using PTWTE-WTM.

>> Temperature-related variables



Conclusion



Conditions
- High Maximum T
- High Running T
- Reasonable Effective sample size

Funding: Cancer Prevention and Research Institute of Texas (CPRIT)